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ORIGINAL ARTICLE

# A novel 3-input XOR function implementation in quantum dot-cellular automata with energy dissipation analysis



Ali Newaz Bahar <sup>a,\*</sup>, Sajjad Waheed <sup>a</sup>, Nazir Hossain <sup>b</sup>, Md. Asaduzzaman <sup>a</sup>

<sup>a</sup> Department of Information and Communication Technology (ICT), Mawlana Bhashani Science and Technology University (MBSTU), Santosh, Tangail 1902, Bangladesh

<sup>b</sup> Department of Electrical and Computer Engineering, University of Massachusetts Lowell, MA 01854, USA

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## KEYWORDS

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**Abstract** Quantum dot-cellular automata (QCA) are one of novel emerging nanotechnology, which seem to be excellent alternatives to the conventional complementary metal-oxide semiconductor (CMOS) technology. QCA technology has a wide range of optimize facet such as ultra-low power consumption, faster switching speed and extremely density structure. In this paper, a novel exclusive-OR (XOR) gate is presented. This proposed XOR gate requires only 12 cells and dissipates 12.11 meV energies at  $1.0E_k$  tunneling energy level. To inspect the efficacy of proposed XOR gate, new QCA design of half and full subtractor is introduced here. In comparison with previous QCA designs, the proposed layouts are implemented with the minimum area, minimum number of cells and delay without any wire-crossing techniques. The proposed, half and full subtractors require 19 and 32 cells and occupy  $0.0186 \mu\text{m}^2$  and  $0.0287 \mu\text{m}^2$  area respectively. To validate the accuracy of the proposed design, QCADesigner, a familiar QCA simulation tool is employed.

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## 1. Introduction

According to Moore's law the numbers of components on a chip will double in every 18 months [1]. In recent times, complementary metal-oxide semiconductor (CMOS) technology

is going to reach its physical limitations and this technology faces several problems in nanoscale designing [2]. Thus, many researchers have been performed to find the alternatives to replace typical CMOS technology. Quantum-dot cellular automata (QCA) are one of the emerging nano technologies, which are promising alternatives to CMOS technology and suitable for designing nanoscale circuits [2,3].

A number of conventional 3-input majority voter and inverter based exclusive-OR gates have been reported in [3–7]. Chabi et al. in [8] first proposed 5-input majority gate based exclusive-OR gate and implemented this gate in multiplexer. A new robust single layer wire crossing 2-input and 3-input XOR has been reported by Angizi et al. in [9].

\* Corresponding author.

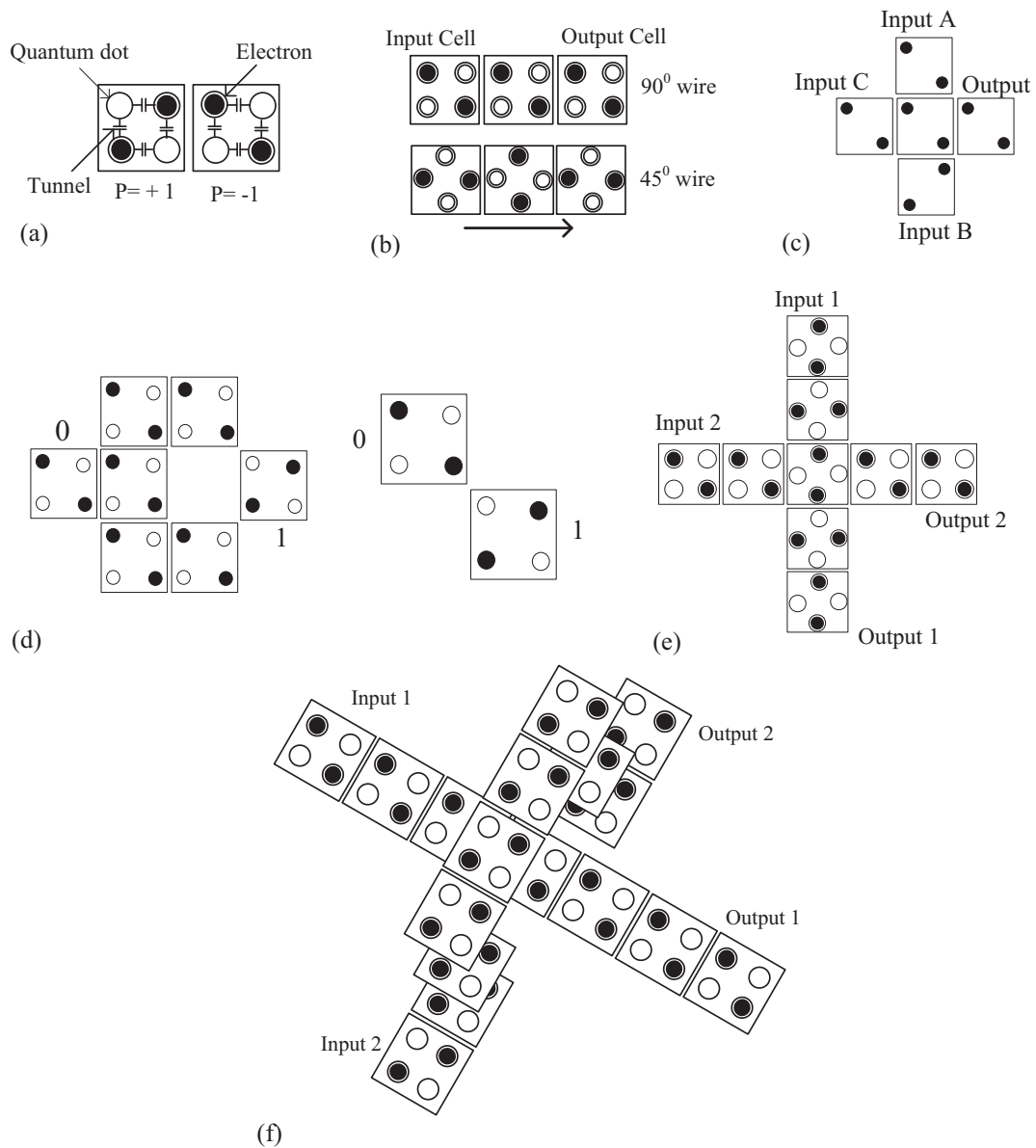
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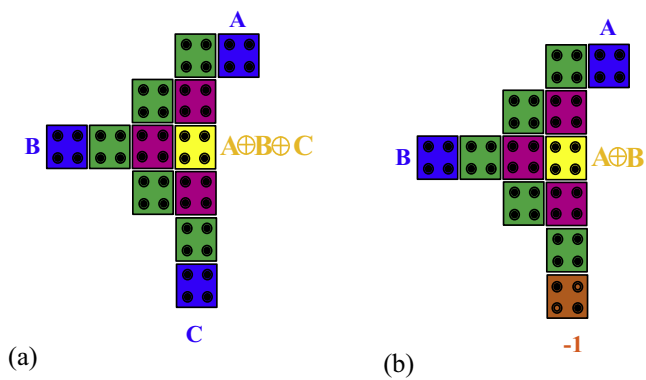
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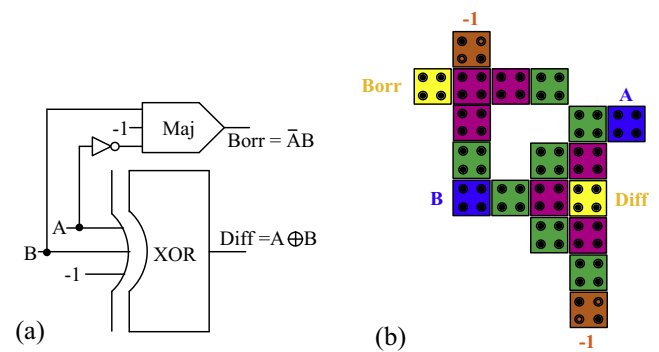
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**Figure 1** (a) QCA cell with two-basis polarization, (b) QCA binary wire, (c) 3-input majority voter gate, (d) different types of QCA inverter, (e) QCA coplanar wire-crossing technique, and (f) QCA multi-layer wire-crossing technique.



**Figure 2** Proposed (a) 3-input exclusive-OR gate, and (b) 2-input exclusive-OR gate.



**Figure 3** (a) QCA block diagram, and (b) QCA representation of proposed half subtractor.

Using low power consumption and dense 5-input majority gate Sheikhfaal et al. [10] have presented a new 2-input XOR gate. In [11] several QCA based XOR gates have been introduced and evaluated the power dissipation at different tunneling energy levels.

Since the first introduced by Lent et al. [5] in 1993, a number of QCA based logic circuits such as memory cells [5,12–15], adder circuit [16–26] and reversible logic circuits [27–33] have been proposed. However, a small number of QCA based binary subtractors have been proposed so far. Fredkin [29] and Feynman [27] gates based different binary subtractors have been proposed in [34]. The design proposed by Lakshmi and Dallaki in [35,36] has required 77 and 55 cells for employing half subtractor. For implementing full subtractor 178 and 136 cells are required in [35,36]. Reshi has proposed an optimized design of half and full subtractor that requires 45 and 104 cells respectively for implementation [37].

In this paper, an innovative design of 2-input and 3-input exclusive-OR gates is introduced. To investigate the effectiveness of our projected exclusive-OR gate, a one bit half and full subtractors are introduced. Those designs are the most optimized designed comparing with all the previous designs. The rest of the paper is organized as follows: the background of QCA technology will be described in Section 2. Section 3, presents the QCA implementation of proposed gate. The simulation and result comparison are discussed in Section 4. In Section 5, power consumption is estimated using the QCAPro tool. The reliability of proposed design is discussed in Section 6. Finally this paper is concluded in Section 7.

## 2. QCA background

The basic component of QCA is the QCA cell which has four quantum dots, and two of them are mobile electrons [3]. The electrons are capable to exchange their position between adjacent dots but not between neighboring cells [3]. The two electrons are located at diagonally to ensure the maximum distance for Coulombic repulsion and thus depending on the electron's position the QCA cell has two types of polarizations: polarization  $P = +1$  (to represent binary "1") and  $P = -1$  (to represent binary "0") shown in Fig. 1(a). When a set of QCA cells are arranged one after another with the same polarization this set of cell arrangement is known as QCA wire. QCA wire can propagate signal from one end to other end shown in Fig. 1(b). In QCA, there are two types of crossovers: one coplanar crossover and other multi-layer crossover as shown

in Fig. 1(e) and (f), respectively. In coplanar crossing signals are passing through the two different types of wire (binary wire and inverter chain) in the same layer, but in multi-layer crossover, signals are passed through the binary wire via more than one layer. All QCA logic circuits can be implemented using majority gate, and inverter gate. The majority gate or voter (MV) is the fundamental logical unit of QCA design. There are different types of majority gate such as 3-input majority gate, 5-input majority gate, 7-input majority gate. But 3-input majority gate is the most effective and frequently used gates which is employed by five cells. This MV has three input cells, one middle cell and one output cell shown in Fig. 1(c). The logic function of the 3-input majority gate is given in Eq. (1).

$$O = MV(A, B, C) = A \cdot B + A \cdot C + B \cdot C \quad (1)$$

where  $A$ ,  $B$  and  $C$  are the inputs, and  $O$  is the output.

The 3-input majority gate can be configured as 2-input AND or 2-input OR gate by simply fixing one of the inputs polarization to  $P = -1$  or  $P = +1$  respectively. For example if we set input  $C$ 's polarization  $P = -1$  (logic "0") then we get AND gate and if we set  $P = +1$  (logic "1") then we get OR gate.

$$A \cdot B = MV(A, B, 0) \quad (2)$$

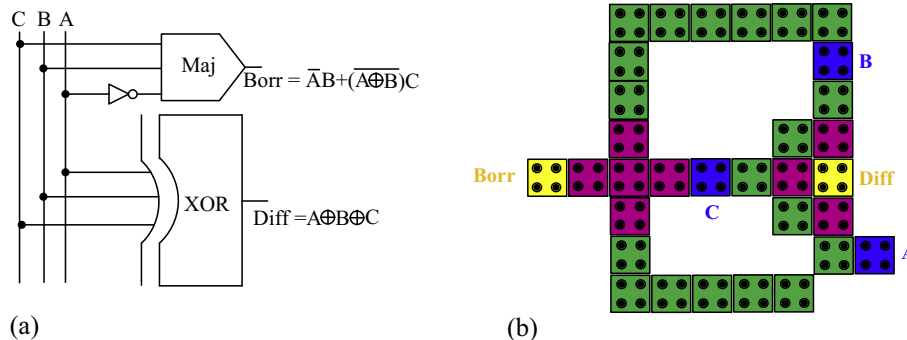
$$A + B = MV(A, B, 1) \quad (3)$$

The QCA inverter returns the opposite polarizations to its input polarization, and different types of inverters are shown in Fig. 1(d).

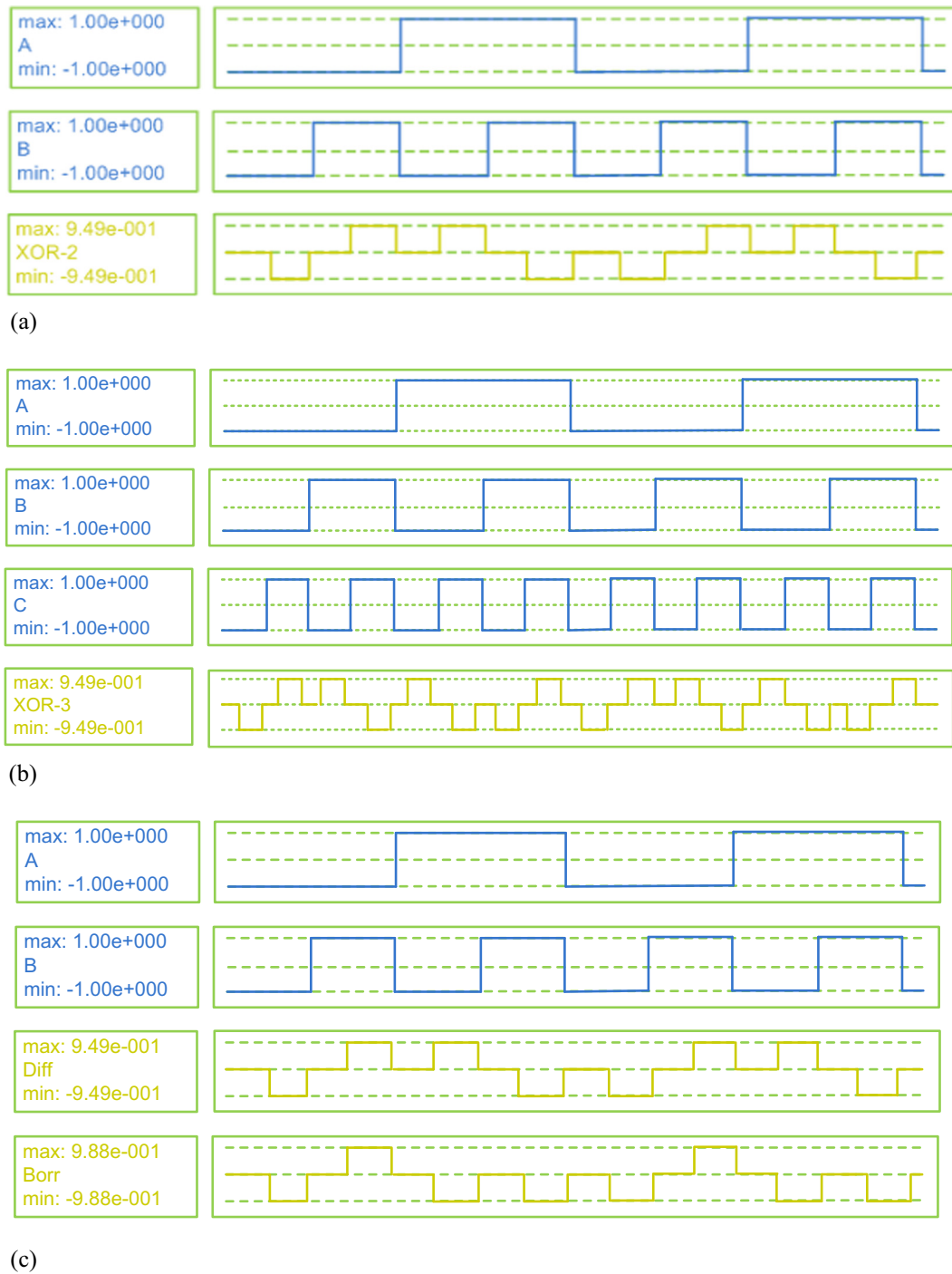
## 3. Proposed gates and QCA implementation

### 3.1. Novel exclusive-OR gate

Exclusive-OR (XOR) gate has a wide range of applications in digital logic circuits such as arithmetic and logical unit, parity checking and detection circuit, reversible logic circuit, code converter and so on. That's why designing an efficient and low power high speedy XOR gate is one of the most exciting research areas in QCA. The conventional and low-complexity exclusive-OR gate has been introduced by different authors [6–9]. But our proposed 3-input exclusive-OR ( $XOR_3$ ) and 2-input exclusive-OR ( $XOR_2$ ) are simple in structure and more efficient compared to the previous design. The significant contribution of this design is implemented in single layer with-



**Figure 4** (a) Schematic block diagram, and (b) QCA representation of the proposed full-subtractor.



**Figure 5** Simulated input-output waveform of proposed (a) 2-input XOR, (b) 3-input XOR, (c) half subtractor, and (d) full subtractor.

out any wire crossing and required only 12 cells, as shown in Fig. 2.

### 3.2. Half subtractor

A subtractor is a combinational circuit and one of the fundamental arithmetic units which carry out the subtraction of the input bits. A half subtractor has two inputs known as minuend

and subtrahends with two outputs called as difference and borrow. For a half subtractor, consider the inputs and outputs are labeled as A, B, Diff (difference) and Borr (Borrow) respectively, and the relationship between the inputs and the outputs is given as follows:

$$Diff = Difference = A \oplus B \quad (4)$$

$$Borr = Borrow = \overline{A}B \quad (5)$$

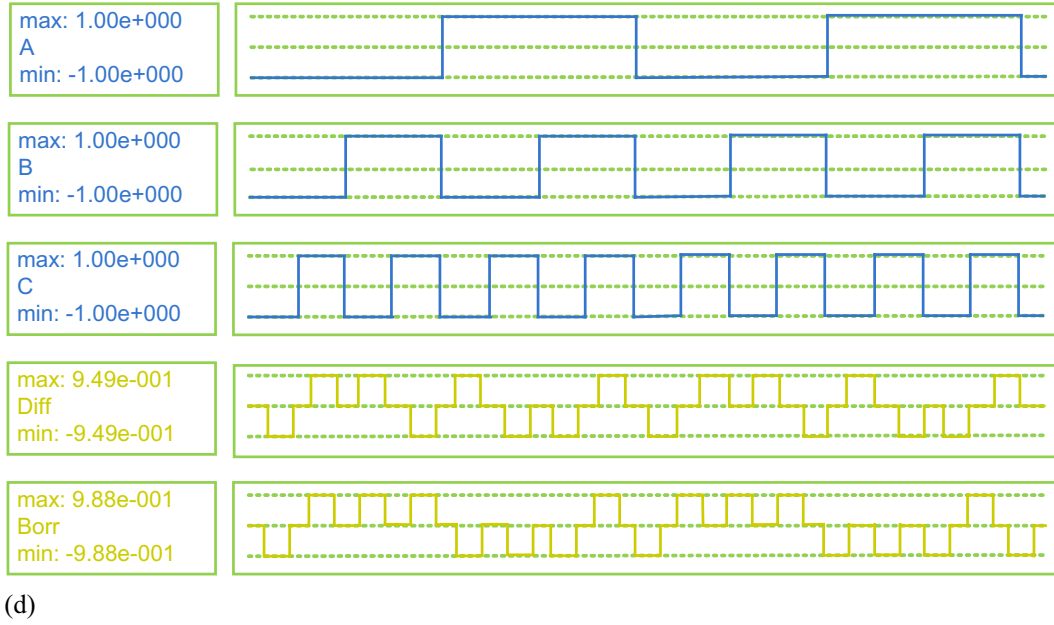


Fig. 5 (continued)

**Table 1** Comparison of proposed exclusive-OR gate and subtractors with the previous works.

| Circuit                  | Number of cells | Approximated area ( $\mu\text{m}^2$ ) | Latency (clock cycle) | Gate count (MV + Inverter) | Type of wire crossing |
|--------------------------|-----------------|---------------------------------------|-----------------------|----------------------------|-----------------------|
| XOR gate [9]             | 67              | 0.06                                  | 1.25                  | 3                          | Coplanar              |
| XOR gate [10]            | 32              | 0.02                                  | 1                     | 3                          | Not required          |
| XOR gate [36]            | 30              | 0.0233                                | 0.75                  | 4                          | Not required          |
| XOR gate [11]            | 28              | 0.02                                  | 0.75                  | 4                          | Not required          |
| Proposed XOR             | 12              | 0.0116                                | 0.50                  | 1                          | Not required          |
| Half subtractor [35]     | 77              | 0.0896                                | 3                     | 7                          | Coplanar              |
| Half subtractor [36]     | 55              | 0.0504                                | 3                     | 6                          | Coplanar              |
| Half subtractor [37]     | 45              | 0.0356                                | 3                     | 6                          | Not required          |
| Proposed half subtractor | 19              | 0.0186                                | 2                     | 3                          | Not required          |
| Full subtractor [35]     | 178             | 0.205                                 | 8                     | 15                         | Coplanar              |
| Full subtractor [36]     | 136             | 0.168                                 | 7                     | 13                         | Coplanar              |
| Full subtractor [37]     | 104             | 0.1043                                | 7                     | 14                         | Not required          |
| Proposed full subtractor | 32              | 0.0287                                | 2                     | 3                          | Not required          |

For implementing half subtractor in QCA one  $\text{XOR}_2$  gate and one AND gate are employed. This proposed layout uses only 19 cells, occupies  $0.0186 \mu\text{m}^2$  area and requires two clock cycles to generate the correct output. Fig. 3 presents the QCA block diagram and circuit layout the proposed half subtractor.

### 3.3. Full subtractor

In full subtractor, previous borrow is considered for calculating the final result. Now, consider a full subtractor with three inputs labeled as A, B, and C and two outputs labeled as Diff and Borr, and then the relationship between inputs and outputs is given as follows:

$$\text{Diff} = \text{Difference} = A \oplus B \oplus C \quad (6)$$

$$\text{Borr} = \text{Borrow} = \overline{A}B + (\overline{A} \oplus B)C \quad (7)$$

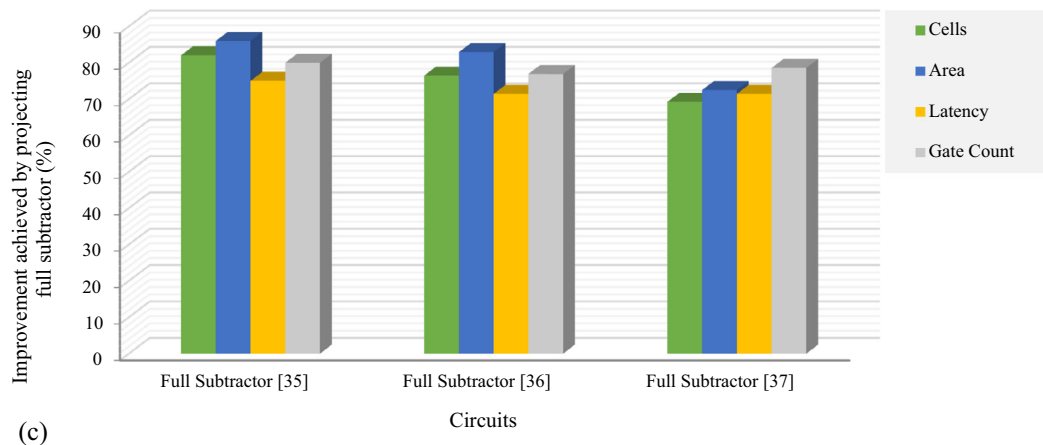
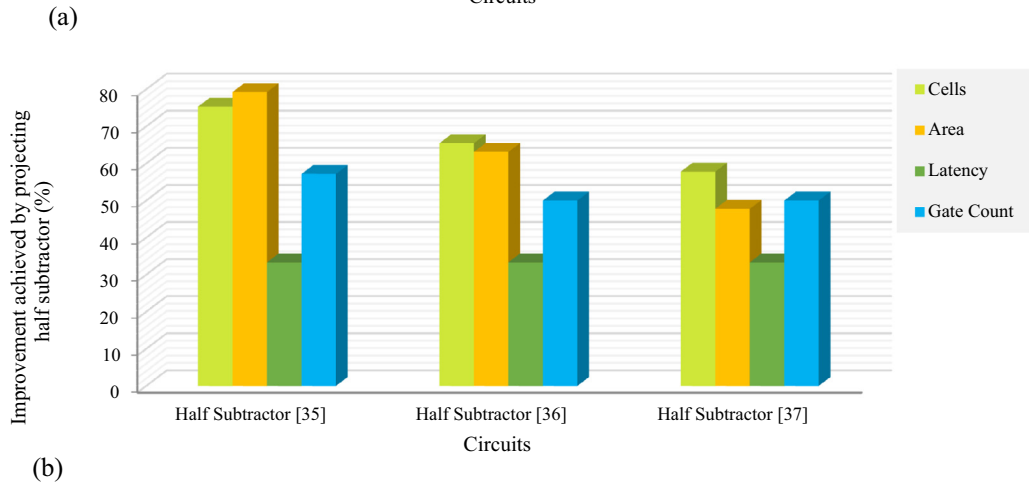
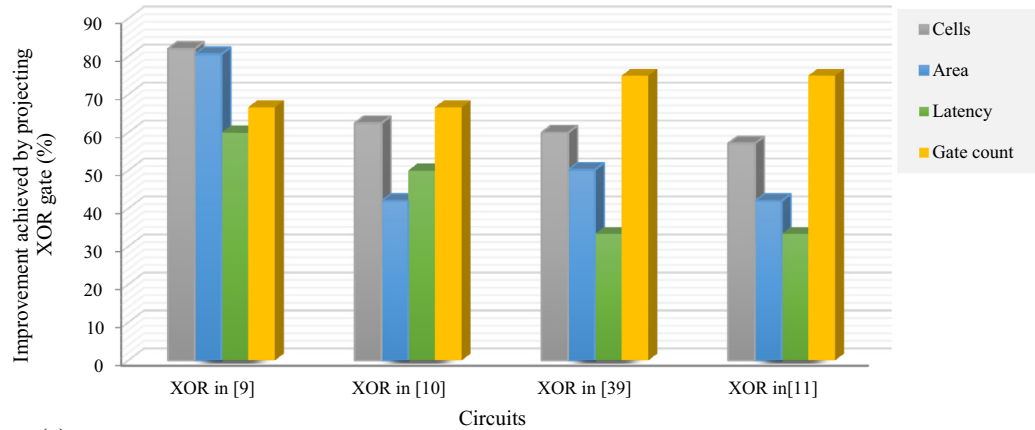
And the QCA based representation of above equation is

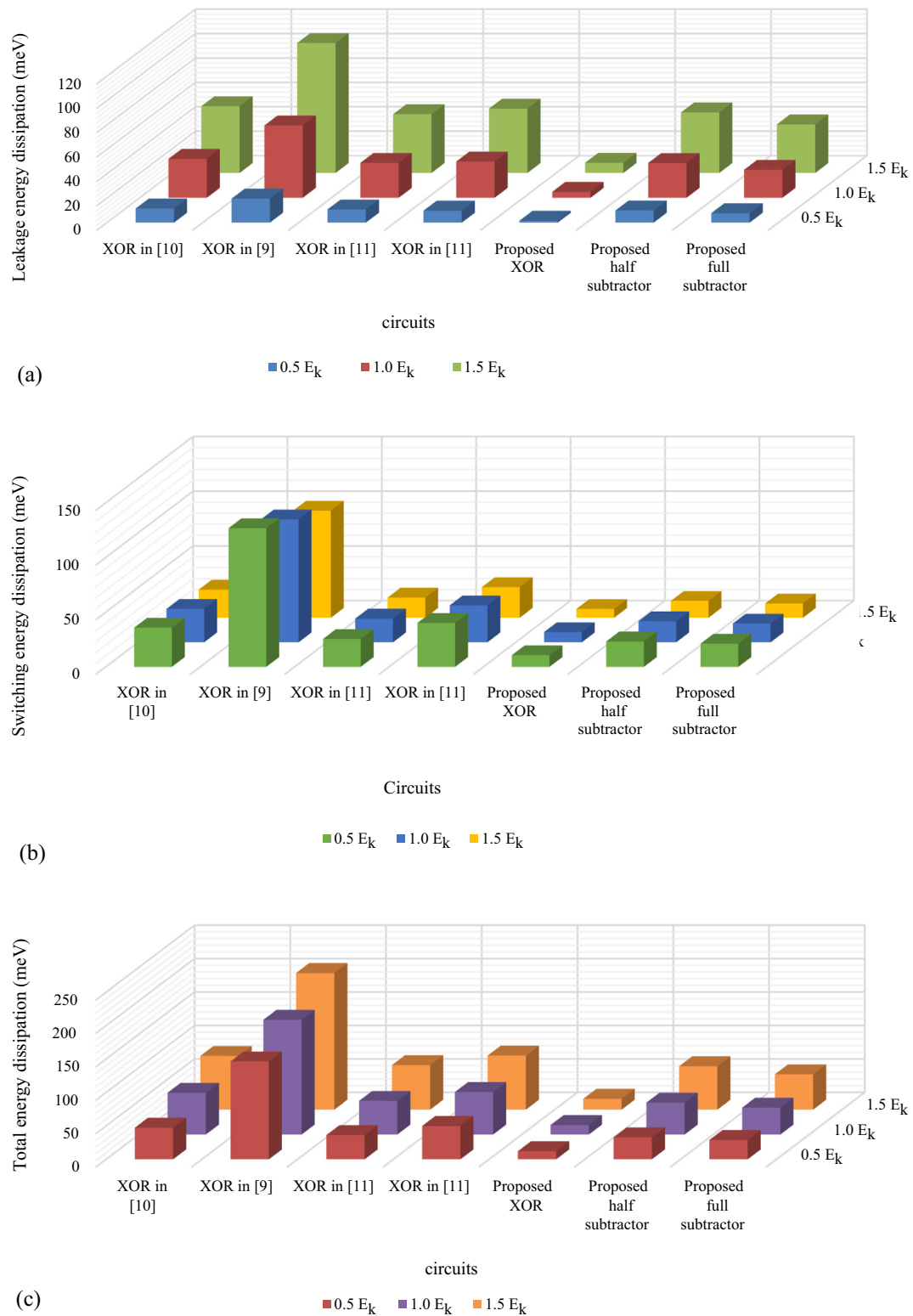
$$\text{Diff} = \text{XOR}_3(A, B, C) \quad (8)$$

$$\begin{aligned}
 \text{Borr} &= \overline{A}B + (\overline{A} \oplus B)C \\
 &= \overline{A}B + (\overline{A}B + AB)C \\
 &= \overline{A}B + \overline{A}BC + ABC \\
 &= \overline{A}B + \overline{A}BC + \overline{A}BC + ABC + \overline{A}BC \\
 &= \overline{A}B + \overline{A}C(\overline{B} + B) + BC(\overline{A} + A) \\
 &= \overline{A}B + BC + \overline{A}C \\
 &= \text{MV}(\overline{A}, B, C)
 \end{aligned} \quad (9)$$

**Table 2** Energy dissipation analysis of XOR gate.

| Circuit                  | Leakage energy dissipation (meV) |          |          | Switching energy dissipation (meV) |          |          | Total energy dissipation (meV) |          |          |
|--------------------------|----------------------------------|----------|----------|------------------------------------|----------|----------|--------------------------------|----------|----------|
|                          | $0.5E_k$                         | $1.0E_k$ | $1.5E_k$ | $0.5E_k$                           | $1.0E_k$ | $1.5E_k$ | $0.5E_k$                       | $1.0E_k$ | $1.5E_k$ |
| XOR in [10]              | 11.51                            | 31.91    | 54.69    | 35.78                              | 30.48    | 25.66    | 47.29                          | 62.39    | 80.34    |
| XOR in [9]               | 19.65                            | 59.28    | 106.51   | 126.79                             | 112.29   | 97.95    | 146.44                         | 171.57   | 204.47   |
| XOR in [11]              | 10.78                            | 28.57    | 48.15    | 25.43                              | 21.71    | 18.4     | 36.20                          | 50.28    | 66.58    |
| XOR in [11]              | 9.73                             | 29.7     | 52.78    | 40.08                              | 33.73    | 28.05    | 49.81                          | 63.49    | 80.83    |
| Proposed XOR             | 1.54                             | 4.85     | 8.27     | 10.58                              | 9.32     | 8.02     | 12.11                          | 14.17    | 16.28    |
| Proposed half subtractor | 9.84                             | 28.47    | 49.61    | 23.15                              | 19.29    | 15.37    | 32.98                          | 47.61    | 64.98    |
| Proposed full subtractor | 7.73                             | 22.93    | 39.81    | 21.22                              | 17.18    | 13.05    | 28.95                          | 40.11    | 52.85    |

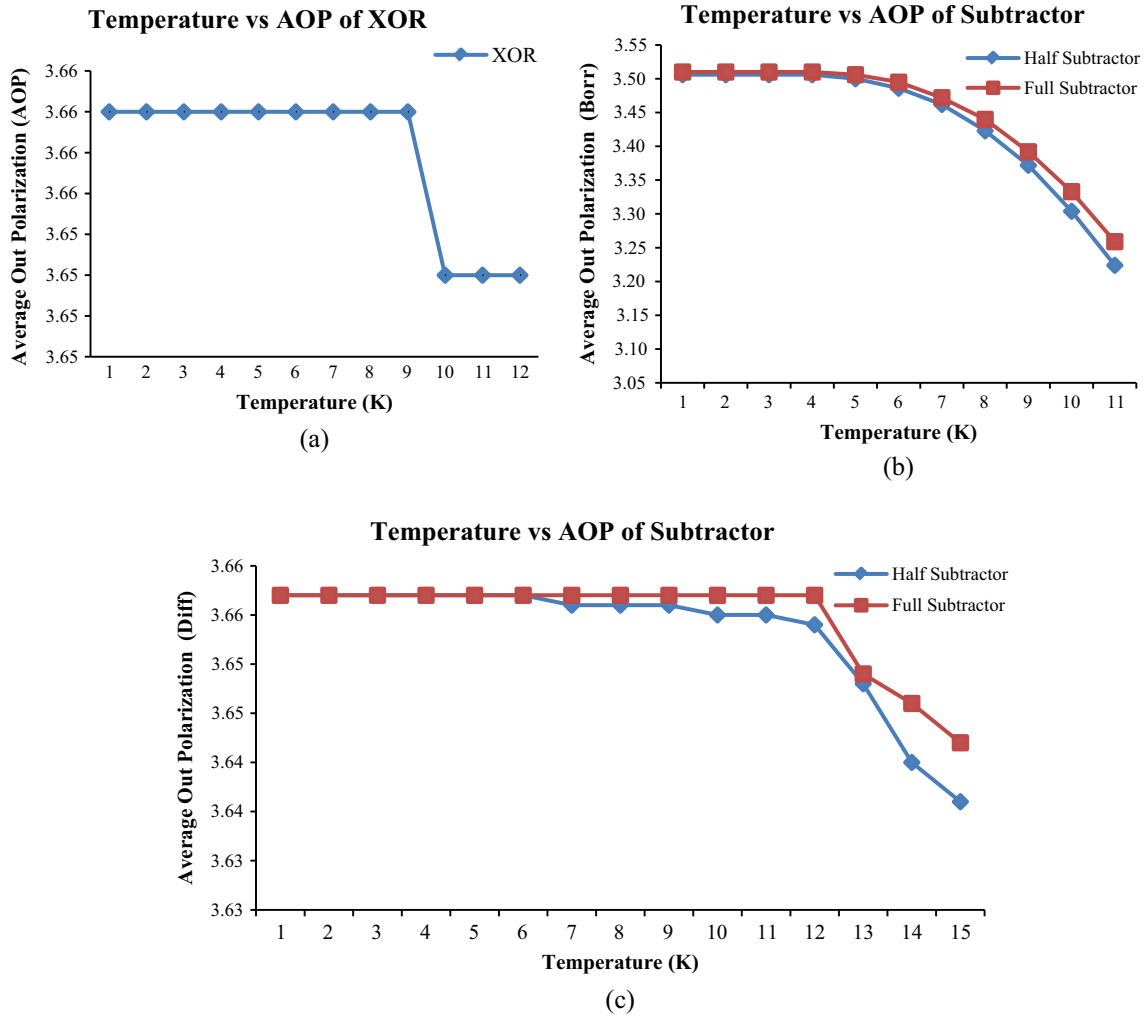
**Figure 6** Overall improvement achieved by proposed (a) exclusive-OR gate, (b) half subtractor, and (c) full subtractor over the previous design.



**Figure 7** The (a) average leakage energy dissipation, (b) average switching energy dissipation and (c) total energy dissipation of the presented XOR, half subtractor and full subtractor at three different tunneling energy levels ( $T = 2.0$  K).

In order to implement the full subtractor in QCA, we use one XOR gate and one 3-input majority gate. In addition, this design requires only 32 cells and needs two clock pulses to pro-

duce accurate output. The schematic block diagram and QCA representation of the proposed full-subtractor are presented in Fig. 4.



**Figure 8** Temperature effect on average output polarization of proposed circuits: (a) exclusive-OR, (b) subtractor for output “Borr”, and (c) subtractor for output “Diff”.

#### 4. Simulation result and discussion

To design and verify the proposed circuits QCADesigner version 2.0.3 [38] is employed with default parameters of the both Bistable and Coherence Vector engines. For the both simulation engines the identical outcomes are achieved, which point toward the accuracy of the proposed design. The simulation results of the proposed 2-input XOR, 3-input XOR, QCA half and full subtractors are shown in Fig. 5. Table 1 demonstrates the comparison between proposed design and previously reported design.

The proposed exclusive-OR gate requires 57% less cells and occupies 42% less area comparing to the previous best design [11] as shown in Fig. 6(a). For designing the half subtractor, we use 58% less cells and the circuit consumes 48% less area comparing to [37]. On the other hand, the proposed full subtractor needs 69% less cells, takes 73% less area and uses 72% less clock push over the earlier best layout [37]. In Fig. 6, the overall improvement of proposed design is shown.

#### 5. Power dissipation analysis

The power dissipation by a QCA cell is calculated using the Hartree-Fock approximation. The Hamiltonian matrix of a mean-field approach is illustrated as [39–41]

$$H = \begin{bmatrix} \frac{-E_k}{2} \sum_i C_i f_{ij} & -\gamma \\ -\gamma & \frac{E_k}{2} \sum_i C_i f_{ij} \end{bmatrix} = \begin{bmatrix} \frac{-E_k}{2} (C_{j-1} + C_{j+1}) & -\gamma \\ -\gamma & \frac{E_k}{2} (C_{j-1} + C_{j+1}) \end{bmatrix} \quad (10)$$

The energy cost of two neighboring cells ( $i$  and  $j$ ) with opposite polarizations is derived as follows:

$$E_{ij} = \frac{1}{4 \prod \epsilon_0 \epsilon_r} \sum_{n=1}^4 \sum_{m=1}^4 \frac{q_{i,n} q_{j,m}}{|r_{i,n} - r_{j,m}|} \quad (11)$$

For every clock cycle the expectation energy value of cell is measured as follows:

$$E = \langle H \rangle = \frac{\hbar}{2} \cdot \vec{\Gamma} \cdot \vec{\lambda} \quad (12)$$

For any instance the total power of a single QCA cell can be calculated as follows:

$$P_t = \frac{dE}{dt} = \frac{\hbar}{2} \left[ \frac{d\vec{\Gamma}}{dt} \cdot \vec{\lambda} \right] + \frac{\hbar}{2} \left[ \vec{\Gamma} \cdot \frac{d\vec{\lambda}}{dt} \right] = P_1 + P_2 \quad (13)$$

In one clock cycle the energy dissipation of a QCA cell is estimated in respect of Hamiltonian and Coherence vectors as follows:

$$E_{diss} = \frac{\hbar}{2} \int_{-T}^T \vec{\Gamma} \cdot \frac{d\vec{\lambda}}{dt} dt = \frac{\hbar}{2} \left( \left[ \vec{\Gamma} \cdot \vec{\lambda} \right]_{-T}^T - \int_{-T}^T \vec{\lambda} \cdot \frac{d\vec{\Gamma}}{dt} dt \right) \quad (14)$$

And the upper bound power dissipation model [37] is given as

$$P_{diss} = \frac{E_{diss}}{T_{cc}} \left\langle \frac{\hbar}{2T_{cc}} \vec{\Gamma}_+ \times \left[ -\frac{\vec{\Gamma}_+}{|\vec{\Gamma}_+|} \tanh \left( \frac{\hbar |\vec{\Gamma}_+|}{k_B T} \right) + \frac{\vec{\Gamma}_-}{|\vec{\Gamma}_-|} \tanh \left( \frac{\hbar |\vec{\Gamma}_-|}{k_B T} \right) \right] \right\rangle \quad (15)$$

In order to calculate the power dissipation of proposed XOR, half subtractor and full subtractor, we employed QCAPro [41]. The power dissipation is measured in different tunneling energy levels at 2 K temperature, shown in Table 2. In addition, the average leakage energy, average switching energy and average energy dissipation at three separate tunneling energy levels  $\gamma = 0.5E_k$ ,  $\gamma = 1E_k$  and  $\gamma = 1.5E_k$  of proposed design and earlier reported designs are illustrated in Fig. 7. According to Table 2, it is worth to mention that the proposed XOR design dissipates 58% less switching energy, 86% less leakage energy and 67% less average energy than previous best design in [11] at  $0.5E_k$  tunneling level.

## 6. Reliability analysis

The temperature effect on average out polarization (AOP) of proposed gates has been analyzed by QCADesigner. The AOP is taken at different temperatures and the effect is shown in Fig. 8. For, XOR gate the AOP remains the same up to 9k and accurate out is generated. When the temperature is above 9k, the AOP falls down radically and produces inconsistent outputs. Similarly, the output “Borr” for both half and full subtractor works perfectly up to 8k. On the other hand, average output polarization for output “Diff” for both half and full subtractor remains the same up to 12k. The “Diff” instigate malfunctioning for both the subtractor circuits at temperature 13k.

## 7. Conclusion

In this paper, an efficient single layer exclusive-OR gate has been proposed. This gate is simple and constructed using conventional cells. In addition, the proposed XOR gate is required 57% less cells and dissipates 71% less power at  $1.0E_k$ , comparing to the previous best design [11]. Moreover, this gate is suitable for digital logic design and the effectiveness of our projected gate has been investigated by implementing in one bit half and full subtractors. Our proposed half subtractors have achieved 58% improvement in cell count and consume 48% less area compared to [37]. On the other hand, full subtractors are 73% area efficient compared to the previous best reported design in [37]. Finally, the results confirmed the dom-

inance of our designs over state-of-the-art designs in terms of power consumption, consumed cell count, area occupation and circuit delay.

## Conflict of interest

The authors declare that they have no conflict of interest.

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